

ULIP Integration Document

Version No.: 5.0

Date: 30/03/2026

Project Name: ULIP

This document contains proprietary information of NLDSL. Unauthorized access, copying and replication are prohibited. This document must not be copied in whole or part by any means, without the written authorization of NLDSL, Noida, India.

Revision History

Version	Date	Author	Reviewer	Significant Changes
1.0	09/07/2021	Hasan Mhad Khan	Nayab Ali	Baseline Version
2.0	09/08/2022	Akshay Singh	Nayab Ali	Update Request Body Regex
3.0	14/03/2024	Ayush Arya	Nayab Ali	Added Fastag 02 API
4.0	01/10/2024	Ayush Arya	Nayab Ali	Updated
5.0	30/03/2026	Gaurav Tyagi	Pranab Manna	Updated

Table of Contents

Introduction	4
1.1 Purpose	4
The Overall Description	4
1.2 Data Integration between FASTAG System and ULIP application	4
1.3 FASTAG/01	4
1.3.1 Technical Approach.....	5
1.4 FASTAG/02	8
1.4.1 Technical Approach.....	8
1.5 Data Transmission & Authentication Mechanism	16

Introduction

1.1 Purpose

The purpose of this document is to define and provide details for accessing API for transaction details from NETC Fastag (National Electronic Toll Collection) system from ULIP system. This document details the following points:

1. **Data Integration between FASTag and ULIP web service:** Identification of operational points where data integration between FASTAG and ULIP web service will be needed.
2. **Technical Approach** for performing Data Integration

The Overall Description

Users will share information such as vehicle registration numbers based on data send by user ULIP system will call **FASTag** API for accessing information.

Following are the points for accessing vehicle information: -

1. This API will share data based on information provided by users.
2. ULIP will expose API for getting data from FASTAG API.
3. **FASTAG/01** API will take vehicle registration number from user and get data from FASTAG API.

1.2 Data Integration between FASTAG System and ULIP application

Users will share information with ULIP system based on data shared ULIP will provide data to user.

1.3 FASTAG/01

This API will take **Vehicle registration number / Vehicle Chassis number** and connect with FASTAG API to get data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	vehiclenumber	Vehicle registration number / Vehicle Chassis number as available in FASTAG system.	^[A-Z0-9]{5,11}\$ ^[A-Z0-9]{17,20}\$	Required	(5 to 11) and (17 to 20)

1.3.1 Technical Approach

1. All the data will be shared through rest web services.
2. ULIP system will provide **vehicle registration number / vehicle chassis number** for which detail is required.
3. FASTAG API will share details of vehicle transaction information which exist in FASTAG system in JSON format.
4. ULIP system will provide the data in JSON Object format in response body.

NOTE: Data related to VRN will be available for past 72 Hours.

Example: the first transaction happens on 1st Jan (data will be available for 3rd Jan, on 4th Jan no data will be available regarding the previous transaction i.e. 1st Jan)

Example:

Vehicle registration number **GA060000** is shared by ULIP system, then FASTAG-integration will make a request to FASTAG API.

Request:

The request must be of the following format, where **vehicle registration number** is passed within the URL body in JSON format.

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/01>

```
{"vehiclenumber": "GA060000"}
```

Curl URL

```
curl --location --request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0liwiaWF0IjoxNjMwNTAyNDI1LCJhcHBzIjoiZGF0YXB1c2gifQ.C7nHJJvr2h7UEufpnDOBUPkHCCCDONtVg5iK5Dw7uEgqxqHYxJkPGnkoNcy02oj7JYHQzDcHfEpbvVenUMdvl_A' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--data-raw '{"vehiclenumber": "GA060000"}'
```

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/01>

```
{"vehiclenumber": "CG07BC9186"}
```

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/01' \
--header 'Authorization: Bearer
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ1bGlwMlslmlhdCI6MTcyNzY5MDAwMCwiYXBwcyI6ImRhdGFwdXNoIn0.2BrY8dCCS_BAH5Y6faXPYAPJEV52bGuZZRtKXD04AOmAUtooBy9TC-wmWVvxyAWsqLR9ifBhYD9xfqGMDPH9_A' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data '{"vehiclenumber": "CG07BC9186"}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the below response -

a) In case of invalid format

1. Invalid vehicle number

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: vehiclenumber. Format
should follow ^[A-Z0-9]{5,11}$|^[A-Z0-9]{17,20}$"
}
```

b) In case of vehicle numbers, it does not exist in FASTAG system

```
{
  "response": [
    {
      "response": {
        "result": "FAILURE",
        "respCode": "000",
        "ts": "2022-02-15T11:34:31",
        "vehicle": {
          "errCode": "740"
        }
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

c) In case vehicle number exists

```
{
  "response": [
    {
      "response": {
        "result": "SUCCESS",
        "respCode": "000",
        "ts": "2021-11-01T19:20:47",
        "vehicle": {
          "errCode": "000",
          "vehltxnList": {
            "totalTagsInMsg": "2",
            "msgNum": "1",
            "totalTagsInresponse": "2",
            "totalMsg": "1",
            "txn": [
              {
                "readerReadTime": "2021-10-30 12:26:09.0",
                "seqNo": "68d47e2d-c10f-4f57-b12a-2dfb547ce5c8",
                "laneDirection": "N",
                "tollPlazaGeocode": "11.0001,11.0001",
                "tollPlazaName": "GMR Chillakallu Toll Plaza",
                "vehicleType": "VC7",
                "vehicleRegNo": "MH19JK3923"
              },
              {
                "readerReadTime": "2021-10-30 12:41:23.0",
                "seqNo": "6361cf5f-8ddd-46dd-9593-0aa0e1fd5780",
                "laneDirection": "N",
                "tollPlazaGeocode": "11.0001,11.0001",
                "tollPlazaName": "GMR Chillakallu Toll Plaza",
                "vehicleType": "VC7",
                "vehicleRegNo": "MH19JK3923"
              }
            ]
          }
        }
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

Status Code 200: vehicle data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: Server is not responding.

1.4 FASTAG/02

This API will take **Vehicle registration number / tagid** and connect with FASTAG API to get data. We have to enter vehiclenumber or tagid to get the data.

S. No.	Field Name	Field Description	Format	Required/Not – Required	Length
1	vehiclenumber	Vehicle registration number / Vehicle Chassis number as available in FASTAG system.	^[A-Z0-9]{5,11}\$ ^[A-Z0-9]{17,20}\$	Optional	(5 to 11) and (17 to 20)
2	tagid	Tag id as available in FASTAG system.	^[A-Z0-9]{0,25}\$	Optional	(0 to 25)

1.4.1 Technical Approach

5. All the data will be shared through rest web services.
6. ULIP system will provide **vehiclenumber or tagid** for which detail is required.
7. FASTAG API will share details of vehicle information which exist in FASTAG system in JSON format.
8. ULIP system will provide the data in JSON Object format in response body.

NOTE: User must enter vehiclenumber or tagid in request body. If user enters both vehiclenumber and tagid in request body, user will get the error response from Fastag API. Users have to enter one of these either vehiclenumber or tagid to get the response.

Example:

Vehicle registration number **MP09HF4987** or tag id **34161FA8203286140F4064E0** is shared by ULIP system, then FASTAG-integration will make a request to FASTAG API.

Request:

The request must be of the following format, where **vehiclenumber** or **tagid** is passed within the URL body in JSON format.

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/02>


```
{"vehiclenumber": "MP09HF4987",  
"tagid": "34161FA8203286140F4064E0"}
```

-

Curl URL

```
curl --location --request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/02' \  
--  
header 'authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwliwiaWF0IjoxNzEwNDZlOTZlLCJhcHBzljoiZGF0YXB1c2gifQ.Bm3DOCwdDJ6z7_0nf4Vpse2LLYh_3ncSriXYmWqWIZzBWnOd9lCMNp1ZglD8nE94SZQUhBPylZdGLr4iNjjw9Q' \  
--header 'accept: application/json' \  
--header 'content-type: application/json' \  
--data-raw '{"vehiclenumber": "MP09HF4987",  
"tagid": "34161FA8203286140F4064E0"}'
```

For staging -

<https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/02>

```
{"vehiclenumber": "",  
"tagid": "34161FA8203286140F4064E0"}
```

-

Curl URL

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/02' \  
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWliOiJ1bGlwMlslmlhdCI6MTcyNzY5MDAwMCwiYXBwcyI6ImRhZGFwdXNoIn0.2BrY8dCCS_BAH5Y6faXPYAPJEV52bGuZZRtKXD04AOmAUtooBy9TC-wmWVvxyAWsqLR9ifBhYD9xfqGMDPH9_A' \  
--header 'Accept: application/json' \  
--header 'Content-Type: application/json' \  
--data '{"vehiclenumber": "",  
"tagid": "34161FA8203286140F4064E0"}'
```

For obtaining authentication token, refer [Section 1.5](#)

Response

To acknowledge the above request, ULIP will send the response below -

1. Invalid vehicle number

```
{  
  "response": null,  
  "error": "true",  
  "code": "400",  
  "message": "Data format failed OR wrong value entered at: vehiclenumber. Format should follow [A-Z]{2}[0-9]{2}[A-Z]{0,5}[0-9]{4}$"  
}
```

2. Invalid tagid number

```
{
  "response": null,
  "error": "true",
  "code": "400",
  "message": "Data format failed OR wrong value entered at: tagid. Format should follow ^[A-Z0-9]{0,25}$"
}
```

3. In case if we enter vehiclenumber or tagid null or empty

```
{
  "response": [
    {
      "response": "vehiclenumber or tagid :must not be Empty or null, Please enter vehiclenumber or tagid",
      "responseStatus": "ERROR"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

4. In case if we enter both vehiclenumber and tagid

```
{
  "response": [
    {
      "response": {
        "result": "FAILURE",
        "successReqCnt": "0",
        "totReqCnt": "1",
        "respCode": "239",
        "ts": "2024-03-15T09:23:19",
        "vehicle": {
          "errCode": "239"
        }
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

5. In case of vehiclenumber or tagid do not exist in FASTAG system

```
{
  "response": [
    {
      "response": {
        "result": "FAILURE",
        "successReqCnt": "0",
        "totReqCnt": "1",
        "respCode": "239",
        "ts": "2024-03-14T14:07:09",
        "vehicle": {
          "errCode": "239"
        }
      },
      "responseStatus": "SUCCESS"
    }
  ],
  "error": "false",
  "code": "200",
  "message": "Success"
}
```

6. In case vehicle number exists

```
{
  "response": [
    {
      "response": {
        "result": "SUCCESS",
        "successReqCnt": "1",
        "totReqCnt": "1",
        "respCode": "000",
        "ts": "2024-03-14T14:07:48",
        "vehicle": {
          "errCode": "000",
          "vehicledetails": [
            {
              "detail": [
                {
                  "name": "TAGID",
                  "value": "34161FA820328972020FB320"
                },
                {
                  "name": "REGNUMBER",
                  "value": "MP09HF4987"
                }
              ]
            }
          ]
        }
      }
    }
  ]
}
```

```
      "name": "TID",
      "value": "E20034120183C2FFEEB86B77"
    },
    {
      "name": "VEHICLECLASS",
      "value": "VC11"
    },
    {
      "name": "TAGSTATUS",
      "value": "I"
    },
    {
      "name": "ISSUEDATE",
      "value": "2017-07-14"
    },
    {
      "name": "EXCCODE",
      "value": "06"
    },
    {
      "name": "BANKID",
      "value": "607417"
    },
    {
      "name": "COMVEHICLE",
      "value": "T"
    }
  ]
},
{
  "detail": [
    {
      "name": "TAGID",
      "value": "34161FA8203289720E14EEA0"
    },
    {
      "name": "REGNUMBER",
      "value": "MP09HF4987"
    },
    {
      "name": "TID",
      "value": "E20034120137FE0006D097ED"
    },
    {
      "name": "VEHICLECLASS",
```

```
        "value": "VC11"
      },
      {
        "name": "TAGSTATUS",
        "value": "I"
      },
      {
        "name": "ISSUEDATE",
        "value": "2019-01-09"
      },
      {
        "name": "EXCCODE",
        "value": "06"
      },
      {
        "name": "BANKID",
        "value": "607417"
      },
      {
        "name": "COMVEHICLE",
        "value": "T"
      }
    ]
  },
  {
    "detail": [
      {
        "name": "TAGID",
        "value": "34161FA82033E764D9F45341"
      },
      {
        "name": "REGNUMBER",
        "value": "MP09HF4987"
      },
      {
        "name": "TID",
        "value": "34161FA82033E764D9F45341"
      },
      {
        "name": "VEHICLECLASS",
        "value": "VC11"
      },
      {
        "name": "TAGSTATUS",
        "value": "A"
      }
    ]
  }
}
```

```
    },
    {
      "name": "ISSUEDATE",
      "value": "2019-10-11"
    },
    {
      "name": "EXCCODE",
      "value": "00"
    },
    {
      "name": "BANKID",
      "value": "652210"
    },
    {
      "name": "COMVEHICLE",
      "value": "T"
    }
  ]
}
],
"responseStatus": "SUCCESS"
}
],
"error": "false",
"code": "200",
"message": "Success"
}
```

7. In case tag id exists

```
{
  "response": [
    {
      "response": {
        "result": "SUCCESS",
        "successReqCnt": "1",
        "totReqCnt": "1",
        "respCode": "000",
        "ts": "2024-03-14T14:09:07",
        "vehicle": {
          "errCode": "000",
          "vehicledetails": [
            {
              "detail": [
```

```
{
  {
    "name": "TAGID",
    "value": "34161FA8203286140F4064E0"
  },
  {
    "name": "REGNUMBER",
    "value": "MAT735701"
  },
  {
    "name": "TID",
    "value": "E2801105200088444B320B00"
  },
  {
    "name": "VEHICLECLASS",
    "value": "VC4"
  },
  {
    "name": "TAGSTATUS",
    "value": "A"
  },
  {
    "name": "ISSUEDATE",
    "value": "2023-07-10"
  },
  {
    "name": "EXCCODE",
    "value": "00"
  },
  {
    "name": "BANKID",
    "value": "606986"
  },
  {
    "name": "COMVEHICLE",
    "value": "F"
  }
]
},
"responseStatus": "SUCCESS"
},
],
"error": "false",
```

```
"code": "200",
"message": "Success"
}
```

Status Code 200: vehicle data response

Status Code 400: Bad request (Invalid json syntax, invalid json key)

Status Code 401 or 403: Unauthenticated or Unauthorized

Status Code 500: Some internal server errors occurred

Status Code 502: Server is not responding.

1.5 Data Transmission & Authentication Mechanism

All data exchange would be done over secure HTTP (HTTPS). Request from ULIP's system would be made in FASTAG system through a data exchange URL that has following form:

For production -

<https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/01>

Body:

```
{"vehiclenumber": "GA060000"}
```

Access to above data exchange URL would be authorized by use of a security mechanism implemented by the ULIP system. It is as follows:

- First time access: It would require a username and password [which would be shared with user beforehand]. Once user hit with valid user id and password, ULIP system will return a basic authorization token to user, using that token user will communicate with ULIP system until tokens will not get expired.

Summary of HTTP Basic Authentication is as follows:

First time when user hits ULIP secured API through username and password for access token as follows-

```
curl --location --request POST 'https://www.ulip.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{"username": "xxxx",
"password": "xxxx@123"}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWIiOiJ0ZXN0MSIsImhhdCI6MTYyNTIzMDQyNywiYXBwcyI6ImRhZGFwdXN0In0.rhumwWJ8BWeeHe9CYmIV1UKyJk-kLHA_ovX5Zi2OwxEv-td-KpxnfbKDI_8FZsOoggqx3AI-Du1y2qMoUD7uzA'
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which

represents user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for vehicle details as follow -

```
curl -X POST \
  https://www.ulip.dpiit.gov.in/ulip/v1.0.0/FASTAG/01 \
  -H 'accept: application/json' \
  -H 'authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWUiOiJ0ZXN0MSIsImhhdCI6MTYyNTIzMDQyNywiYXBwcyI6ImRhdGFwdXN0Ln0.rhumwWJ8BWeeHe9CYmIV1UKyJk-kLHA_ovX5Zi2OwxEv-td-KpxnfbKDI_8FZsOoggqx3AI-Du1y2qMoUD7uzA' \
  -H 'cache-control: no-cache' \
  -H 'content-type: application/json' \
  -H 'postman-token: 8e8a19a4-ea1e-6373-f616-e2e4af9a1338' \
  -d '{"vehiclenumber": "GA060000"}'
```

For staging –

```
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/user/login' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data-raw '{ "username": "xxxx",
  "password": "xxxx@123"}'
```

Authorization:

Bearer

```
eyJhbGciOiJIUzUxMiJ9.eyJzdWUiOiJ1bGlwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg
```

Upon successful authentication, the user would be logged into the ULIP system and get an access token. Though this token user can communicate with ULIP. This token has expiry time which represents user session. The session time out for the same to be fixed hour, generally it is for 30 minutes. If there is no request coming from user for 30 minutes, then session (token) will be expired, and user need to re authenticate.

Using The above token, user can request ULIP system for details as follow -

```
https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/01
curl --location 'https://www.ulipstaging.dpiit.gov.in/ulip/v1.0.0/FASTAG/01' \
--header 'Authorization: Bearer eyJhbGciOiJIUzUxMiJ9.eyJzdWUiOiJ1bGlwIiwiaWF0IjoxNzI3MTYxMDgwLCJhcHBzIjoizGF0YXB1c2gifQ.XhVZm7NiCD05-v9091WJQhYzdk1a0Shs_ley0T4R2MX63swRyxHbHxh6bVXtzqWvWiSLwwNGRQIcqGhr49hpGg' \
--header 'Accept: application/json' \
--header 'Content-Type: application/json' \
--data '{"vehiclenumber": "CG07BC9186"}'
```